Heart Disease Prediction: An Applied Case Study

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Abstract

This project focuses on predicting the presence of heart disease using machine learning techniques. Initially, a comprehensive review of available datasets, including the UCI Cleveland dataset and insights from existing research, highlighted challenges related to data quality and consistency. The primary analysis centered on the UCI Cleveland dataset, involving meticulous data cleaning, exploratory data analysis (EDA), feature engineering, and principal component analysis (PCA). Several classification models, including Naive Bayes, Random Forest (with various feature engineering attempts), and a Multi-Layer Perceptron (MLP), were implemented and evaluated for binary classification (presence vs. absence of heart disease). Model interpretability was explored using SHAP. An optional extension involved analyzing the larger Sulianova cardiovascular disease dataset to compare model performance in a different data context. The study identifies effective models for heart disease prediction and underscores the importance of data quality and iterative model development.

Contents

1 Introduction: Project Premise

Our project aims to predict the presence of heart disease using various patient features, following the guideline which specifies the target variable "goal" as an integer value from 0 (no presence) to 4. As data science students, we understand that the foundation of any reliable model is high-quality, well-understood data. Therefore, our initial step was to investigate the available datasets for this task.

A common source for heart disease data is the UCI Machine Learning Repository, which contains several datasets like Cleveland, Hungarian, Switzerland, and Long Beach VA. These originally had 76 attributes, but most analyses, including the widely cited ones, focus on a subset of 14 due to extensive missing data in the others.

However, research, notably the thesis by Brandon Simmons (2021) [1], has uncovered significant issues within these commonly used datasets. Simmons investigated datasets from UCI, Kaggle, and Dataport, finding problems related to human errors in encoding, data duplication, and potentially misleading information stemming from these inconsistencies.

Key Findings from Simmons' Investigation

- **Kaggle vs. Cleveland Confusion:** A popular heart disease dataset on Kaggle (often derived from UCI) was found to be closely related to the Cleveland dataset (303 observations). However, Simmons identified critical discrepancies, including imputed missing values and, crucially, a reversed binary target variable encoding in the Kaggle version, leading to potentially misleading conclusions in many public analyses.
- **Statlog Dataset:** Another UCI dataset, Statlog (270 observations), was found by Simmons to be a proper subset of the Cleveland dataset, highlighting risks of duplication if both were used without care.
- **Combined Datasets (e.g., ORIG/MISS):** Attempts to combine multiple heart disease datasets suffered from high duplication rates and potential data quality issues in non-Cleveland/Statlog portions.
- **Recommendation:** Simmons concluded that the Cleveland dataset is the most reliable starting point, provided its few missing values are handled. He recommended recoding its 5-level target to binary (0 = Absence, 1 = Presence) and identified 'thalach', 'oldpeak', and 'cp' as key predictors.

Other Datasets Considered

We also reviewed other specific datasets:

- The **Fedesoriano - Heart Failure Prediction Dataset** on Kaggle [2] (918 observations, 11 features from 5 combined sources) presents similar duplication and quality concerns.
- The **Sulianova - Cardiovascular Disease Dataset** on Kaggle [3] (70,000 records) is much larger, with different features focused on general cardiovascular risk.

Our Approach

Given these insights, our primary analysis focused on the UCI Cleveland dataset. Careful preprocessing, including handling missing values, ensuring correct data types, and deciding on the target variable representation (original 0-4 multi-class vs. binary 0/1), was paramount. The Sulianova dataset was explored as an optional extension.

2 Data Understanding and Preparation (Cleveland Dataset)

The initial phase involved loading and inspecting the four heart disease datasets from UCI: Cleveland, Hungarian, Switzerland, and VA. The Cleveland dataset was chosen for the primary analysis due to its relatively complete nature, as identified in prior research.

2.1 Initial Data Loading and Cleaning

The datasets were loaded using predefined column names: age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, and num (target). Missing values, represented by '?', were converted to NaN. All columns were then converted to numeric types. The Cleveland dataset initially had 303 rows. After identifying and removing rows with any missing values (6 rows in total, affecting 'ca' and 'thal'), the cleaned Cleveland dataset comprised 297 rows and 14 columns. Duplicate rows were checked across all datasets; the Cleveland dataset had no duplicates after initial loading, while others did, which were subsequently removed for consistency if those datasets were to be further analyzed (though the focus remained on Cleveland).

2.2 Feature Remapping

The 'thal' (Thallium stress test result) feature in the cleaned Cleveland dataset was remapped for clinical relevance and modeling suitability: normal (original value 3) was mapped to 0, reversible defect (original value 7) to 1, and fixed defect (original value 6) to 2.

3 Exploratory Data Analysis (EDA) - Cleveland

3.1 Feature Distributions

Distributions of all 13 predictive features in cleveland-dataset were visualized using histograms for continuous features and bar plots for categorical ones. This helped understand the spread, central tendency, and potential imbalances for each feature. (Placeholder for combined distribution plot: `fig_feature_distributions_cleveland.png`)

3.2 Correlation Analysis

A correlation matrix heatmap was generated for all features in cleveland-dataset.

- **Correlation with Target ('num'):** Features showing the strongest positive linear correlation with disease severity ('num') were 'ca' (number of major vessels, r=0.52), 'oldpeak' (ST depression, r=0.50), and the remapped 'thal' (r=0.46). 'thalach' (max heart rate) showed a moderate negative correlation (r=-0.42). Features like 'fbs' and 'chol' had negligible linear correlation with 'num'. (Placeholder for target correlation bar plot: `fig_target_correlation_cleveland.png`)
- **Inter-feature Correlation:** Strong correlations were noted between 'oldpeak' and 'slope' (r=0.58). 'thalach' was negatively correlated with 'slope', 'exang', and 'oldpeak'. (Placeholder for correlation heatmap: `fig_correlation_heatmap_cleveland.png`)

3.3 Outlier Analysis: Cholesterol ('chol')

The 'chol' feature was examined for outliers. While some high values were present (e.g., 1 patient with chol ¿ 500 mg/dL, 4 patients ¿ 400 mg/dL), a boxplot indicated these were few and considered clinically plausible extreme values rather than errors. Thus, they were retained. (Placeholder for cholesterol boxplot: `fig_cholesterol_boxplot_cleveland.png`)

4 Principal Component Analysis (PCA) - Cleveland

PCA was applied to the 13 scaled predictive features to visualize the data in lower dimensions.

- **Multi-class Target (‘num‘ 0-4):** Both 2D and 3D PCA plots showed significant overlap between the 5 classes, with class 0 (no disease) being somewhat more distinct but no clear separation for classes 1-4. (Placeholders: `fig_pca_2d_multiclass.png`, `fig_pca_3d_multiclass.png`)
- **Binary Target (0 vs 1+):** A 3D PCA plot using a binary target (0 for no disease, 1 for any disease) showed a slightly clearer tendency for separation, but substantial overlap remained. (Placeholder: `fig_pca_3d_binary.png`)
- **Cumulative Explained Variance:** The first two principal components explained 24.49% of the variance, and the first three explained 34.66%. To explain at least 80% of the variance, 9 components were needed; for 90%, 10 components; and for 95%, 11 components. This indicates that a few components do not capture the majority of the data's variance. (Placeholder: `fig_pca_cumulative_variance_cleveland.png`)

5 Modeling Strategy and Binary Classification (Cleveland)

Initial attempts with Random Forest on the multi-class target (‘num‘ 0-4) showed poor performance, especially for minority classes (severe disease stages), and signs of overfitting. Consequently, the primary modeling strategy shifted to **binary classification**: predicting presence (1, if original ‘num‘ ¿ 0) versus absence (0, if original ‘num‘ == 0) of heart disease. Data was split into 80% training and 20% testing sets, stratified by the binary target. Numerical features (‘age‘, ‘trestbps‘, ‘chol‘, ‘thalach‘, ‘oldpeak‘, and later engineered numerical features) were standardized using ‘StandardScaler‘.

5.1 Model Implementation and Evaluation (Binary Target)

A helper function was created to evaluate binary classifiers, reporting accuracy, precision, recall, F1-score (for the 'disease' class), AUC, and plotting confusion matrices and ROC curves for the test set. It also reported training accuracy.

5.1.1 Naive Bayes (GaussianNB)

A Gaussian Naive Bayes model was trained on the scaled baseline features.

- Test Accuracy: 90.00%
- AUC: 0.9375
- F1-score (Disease): 0.8889
- False Negatives (FN): 4
- (Placeholder for GNB Confusion Matrix: `fig_cm_gnb_cleveland.png`, ROC: `fig_roc_gnb_cleveland.png`)

5.1.2 Random Forest (Baseline)

A Random Forest classifier (100 estimators, balanced class weight) was trained on the scaled baseline features.

- Test Accuracy: 86.67%
- AUC: 0.9408
- F1-score (Disease): 0.8525
- False Negatives (FN): 5
- (Placeholder for RF Baseline Confusion Matrix: `fig_cm_rf_baseline_cleveland.png`, ROC: `fig_roc_rf_baseline_cleveland.png`)

5.1.3 Feature Engineering Iterations with Random Forest

Several feature engineering techniques were explored, with a Random Forest model re-evaluated at each step:

1. **Ratio Features:** 'chol-per-age', 'tachycardia-index', 'bp-stress-ratio' were created. The RF model with these features showed slightly worse performance (Test Acc: 81.67%, AUC: 0.9401, FN: 8) compared to the baseline RF.
2. **Selected Features (7 features):** Based on statistical tests (ANOVA F-value, Chi-squared) and permutation importance (from RF with engineered features), a subset of 7 features ('thal', 'trestbps', 'ca', 'bp-stress-ratio', 'oldpeak', 'thalach', 'exang') was selected. The RF model with these features performed comparably to the baseline RF but with a slight dip (Test Acc: 85.00%, AUC: 0.9158, FN: 6).
3. **Polynomial and Interaction Features (Degree 2):** Systematically generating these from original numerical features (plus ratio features) led to worse RF performance (Test Acc: 81.67%, AUC: 0.9263, FN: 8) and signs of overfitting.
4. **Discretized (Binned) Features:** Binning 'oldpeak' and 'bp-stress-ratio' and using them with other categorical/engineered features significantly reduced RF performance (Test Acc: 75.00%, AUC: 0.8710, FN: 9).

These feature engineering experiments generally did not improve upon the baseline Random Forest or Naive Bayes models for the Cleveland dataset, often increasing complexity without better generalization or even degrading performance.

5.1.4 Neural Network (MLP with PyTorch)

A simple MLP (2 hidden layers with 64 and 32 neurons, ReLU, Batch Normalization, Dropout) was trained using PyTorch.

- **MLP with Baseline Features:** Test Accuracy: 83.33%, AUC: 0.9447, F1 (Disease): 0.8148, FN: 6.
- **MLP with Selected Features (7 features):** Test Accuracy: 83.33%, AUC: 0.9592, F1 (Disease): 0.7925, FN: 9. While achieving the highest AUC, its recall was lower, leading to more FNs.
- (Placeholders: `fig_mlp_loss.png`, `fig_cm_mlp_cleveland.png`, `fig_roc_mlp_cleveland.png`)

5.2 SHAP Analysis (Random Forest Baseline)

SHAP was applied to the baseline Random Forest model to understand feature contributions.

- **Global Importance:** The SHAP summary plot highlighted 'cp', 'thal', 'ca', 'oldpeak', and 'thalach' as highly impactful features globally. (Placeholder: `fig_shap_summary_cleveland.png`)
- **Local Explanations:** Force plots for individual True Positive, False Negative, and False Positive cases illustrated how specific feature values pushed predictions for those instances. (Placeholders: `fig_shap_tp.png`, `fig_shap_fn.png`, `fig_shap_fp.png`)
- **Misclassifications:** Analysis of mean absolute SHAP values for FPs and FNs indicated which features were most influential in these incorrect predictions. For example, 'cp', 'ca', and 'thal' often had high importance in misclassified cases. (Placeholders: `fig_shap_fp_contrib.png`, `fig_shap_fn_contrib.png`)

5.3 Model Comparison (Cleveland - Binary)

(Placeholder for combined ROC curve plot for Cleveland models: `fig_roc_combined_cleveland.png`)

For the Cleveland dataset, GaussianNB provided the highest test accuracy and the lowest number of false negatives. The baseline Random Forest also performed well, particularly in AUC. The MLP with selected features achieved the highest AUC but had more false negatives. Feature engineering attempts did not consistently improve upon these.

Table 1: Performance Summary for Binary Classification Models on Cleveland Dataset.

Model	Test Acc.	AUC	Precision (Dis.)	Recall (Dis.)	F1-score (Dis.)	FN
GaussianNB	0.900	0.938	0.923	0.857	0.889	4
RandomForest (Baseline)	0.867	0.941	0.885	0.821	0.852	5
RandomForest (Eng. Features)	0.817	0.940	0.870	0.714	0.784	8
RandomForest (Selected Features)	0.850	0.916	0.880	0.786	0.830	6
RandomForest (Poly & Interactions)	0.817	0.926	0.870	0.714	0.784	8
RandomForest (Binned Features)	0.750	0.871	0.760	0.679	0.717	9
PyTorch MLP (Baseline Feats)	0.833	0.945	0.846	0.786	0.815	6
PyTorch MLP (Selected Feats)	0.833	0.959	0.950	0.679	0.792	9

6 Optional Extension: Analysis of Sulianova Dataset

A larger cardiovascular disease dataset from Kaggle (Sulianova, 70,000 records) with different features (e.g., age in days, height, weight, systolic/diastolic BP, lifestyle factors) was explored.

6.1 Data Cleaning and Preprocessing (Sulianova)

The ‘id‘ column was dropped. Age was converted from days to years. Significant cleaning was performed for ‘ap-hi‘ (systolic) and ‘ap-lo‘ (diastolic) blood pressure, removing rows where ‘ap-lo ¿ ap-hi‘ or values were outside plausible physiological ranges (e.g., ap_hi 60-250, ap_lo 30-150). Outliers in height and weight were handled using percentile-based filtering (0.5th and 99.5th percentiles). A BMI feature was engineered. The final processed dataset for Sulianova contained approximately 60,000-65,000 records (exact number depends on filtering effects described in notebook).

6.2 EDA and Visualization (Sulianova)

Feature distributions were plotted. The target variable ‘cardio‘ was well-balanced.

- **PCA:** 2D PCA explained 40.2% of variance, and 3D PCA explained 50.6%. Both showed significant class overlap. Cumulative variance analysis showed 7 components were needed for 80% variance, and 9 for 90%. (Placeholders: `fig_pca_2d_sulianova.png`, `fig_pca_3d_sulianova.png`, `fig_pca_cumulative_sulianova.png`)
- **t-SNE:** Applied to a sample of 5000 data points (perplexity=30), t-SNE revealed several distinct visual clusters in 2D and 3D, but these clusters still contained a significant mixture of both ‘cardio‘ classes, indicating no clear non-linear separation. (Placeholders: `fig_tsne_2d_sulianova.png`, `fig_tsne_3d_sulianova.png`)

6.3 Model Implementation and Evaluation (Sulianova)

A preprocessing pipeline (StandardScaler for numerical, OneHotEncoder for categorical) was used. Models were trained on a 75%/25% train/test split. (Placeholder for combined ROC curve plot for Sulianova models: `fig_roc_combined_sulianova.png`)

Table 2: Performance Summary for Models on Sulianova Dataset.

Model	Test Acc.	AUC	Precision (Dis.)	Recall (Dis.)	F1-score (Dis.)	FN
GaussianNB (Sulianova)	0.697	0.769	0.744	0.591	0.658	3418
RandomForest (Sulianova)	0.713	0.770	0.715	0.697	0.706	2529
XGBoost (Sulianova)	0.734	0.800	0.753	0.689	0.719	2596
MLPClassifier (Sulianova)	**0.735**	**0.805**	**0.764**	0.681	**0.712**	2667

For the Sulianova dataset, MLPClassifier and XGBoost were the top performers, with MLPClassifier achieving the highest test accuracy, AUC, and precision. RandomForest showed signs of overfitting.

7 Overall Conclusion and Future Work

This project successfully implemented and evaluated various machine learning models for heart disease prediction on two distinct datasets.

For the **UCI Cleveland dataset**, simpler models like **Gaussian Naive Bayes** (Test Acc: 90.0%, AUC: 0.938, FN: 4) and a **baseline Random Forest** (Test Acc: 86.7%, AUC: 0.941, FN: 5) demonstrated strong and balanced performance for binary classification. Iterative feature engineering attempts did not consistently yield improvements over these baseline models, sometimes degrading performance or increasing overfitting. SHAP analysis provided valuable insights into feature contributions for the Random Forest model.

For the larger **Sulianova dataset**, more complex models like **MLPClassifier** (Test Acc: 0.735, AUC: 0.805) and **XGBoost** (Test Acc: 0.734, AUC: 0.800) performed best, showcasing their ability to handle larger, potentially more complex data. Visualization techniques like PCA and t-SNE highlighted the inherent class overlap in this dataset.

The key takeaways include the critical importance of thorough data understanding and preprocessing, the context-dependent success of feature engineering, and the realization that more complex models are not always superior, especially on smaller datasets. The choice of the "best" model also depends on the specific priorities (e.g., minimizing false negatives vs. maximizing overall accuracy).

Future Work could involve:

- Extensive hyperparameter tuning for all models, especially Random Forest, XGBoost, and MLP.
- Exploring other advanced classifiers (e.g., Support Vector Machines with different kernels, LightGBM).
- More sophisticated feature selection techniques (e.g., Recursive Feature Elimination).
- Investigating ensemble methods further by combining predictions from diverse models.
- Deeper analysis of misclassified instances to understand patterns and potential data issues.
- Addressing ethical implications and fairness in predictive healthcare models.

8 Student Contributions

- **Lucchina Luca**
  - Dataset review and data quality issues (Cleveland, Kaggle, etc.)
  - Research and summary of Simmons' findings
  - Writing: *Introduction*, *Data Understanding*
  - SHAP analysis (global and local)
- **Nazifi Endrit**
  - Cleaning and preprocessing of the Cleveland dataset
  - EDA: distributions, correlations, outliers
  - PCA: analysis, visualizations, interpretation
  - Support in model evaluation
  - Feature engineering (ratios, polynomial features, selection)
- **Rava Miro**
  - Experiments with Random Forest and MLP models
  - Performance visualization (ROC curves, confusion matrix)
  - Writing: *Modeling Strategy*
  - Implementation: Naive Bayes models
- **Del Grande Matteo**
  - Full analysis of the Sulianova dataset
  - PCA and t-SNE on the extended dataset
  - Writing: *Optional Extension*, *Conclusion and Future Work*
  - Implementation of global models
- **Note**
  - All group members contributed jointly to the implementation, training, and evaluation of the classification models (Naive Bayes, Random Forest, MLP, XGBoost) on both datasets.

References

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[2] Fedesoriano. (n.d.). *Heart Failure Prediction Dataset*. Kaggle. Available at: https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

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[4] Dua, D. and Graff, C. (2019). *UCI Machine Learning Repository* [http://archive.ics.uci.edu/ml]. Irvine, CA: University of California, School of Information and Computer Science. (General reference for UCI datasets).